

# Biotechnology and Food Engineering

## Process Biotechnology

### Spring 2026

#### Tutorial #4 – sterilization

- Bacterial death by heat can be described as a first-order chemical reaction:

$$\frac{dN}{dt} = -K_d \cdot N$$

$N$  – number of active bacteria

$t_{st}$  – sterilization period

$K_d$  – sterilization constant (specific death rate)

- After integration:

$$\ln \frac{N_f}{N_0} = -K_d \cdot t_{st}$$

$N_0$  - number of active bacteria at the beginning of sterilization.

$N_f$  - number of active bacteria remaining after sterilization (the probability to find active bacteria after sterilization).

Or: 
$$\frac{N_f}{N_0} = e^{-K_d \cdot t}$$

**$K_d$  depend on temperature by Arrhenius equation:  $K_d = A \cdot e^{-E/RT}$**

$A$  – constant (time<sup>-1</sup>)

$E$  – activation energy (cal/mole)

$R$  – universal gas constant

$T$  – absolute temperature (°K)

### Decomposition of medium components

- It is reasonable to assume that spoilage of the medium components as a result of the heat acts as first-order reaction as well:

$$\ln \frac{C_f}{C_0} = -K' \cdot t$$

$C_0$  – the component amount in the medium in time  $t=0$

$C_f$  – the component amount in the medium after sterilization.

$K'$  – decomposition constant of a specific component (e.g. vitamin)

### Continuous heat exchanger

- The staying time in continuous heat exchanger is bigger or equal to the sterilization period (in case when the heating and cooling periods are negligible, the staying time is equal to the sterilization period).

❖ The staying time ( $t$ ) can be calculated by:

$$t = \frac{L}{U}$$

Where:  $U$  is the flow velocity and  $L$  is the sterilizer length.

$$\ln \frac{N_0}{N_f} = K \cdot t = K \cdot \frac{L}{U} \quad \rightarrow \quad L = \frac{U}{K} \cdot \ln \frac{N_0}{N_f}$$

Flow velocity in heat exchanger,  $U$ :

$$U = \frac{Q \text{ (flow rate of medium/air)}}{A \text{ (cross section area of the heat exchanger)}}$$

❖ The staying time ( $t$ ) can also be calculated by:

$$t = \frac{V_{st}}{Q}$$

Where:  $Q$  is the flow rate of medium/air and  $V_{st}$  is the sterilizer volume.

### Additional terms

- Level of sterilization –  $\frac{N_f}{N_0}$
- Magnitudes –  $\log\left(\frac{N_f}{N_0}\right)$

### **Question 1**

A factory produces protein which is secreted by a certain yeast, in a production fermenter with the volume of  $1 \text{ m}^3$ . The carbon source in the fermentation is provided by a molasses (65% solids) with initial concentration of 30%, and the initial contamination of the molasses is  $10^4$  bacteria per gram. Additionally, the medium contains 15 gr/lit salts with an initial contamination of  $10^6$  bacteria per kilogram, and water which completes the fermenter volume and contains initial contamination of  $2 \cdot 10^3$  bacteria per milliliter.

The medium is sterilized at a temperature of  $132^\circ\text{C}$  ( $K_d=1.75 \text{ min}^{-1}$ ), then it is being inoculated by 400 gr yeast and the fermentation immediately starts with logarithmic growth and ends when the yeast concentration in the fermenter reaches to 2.2%. The fermentation requirements enable bacterial contamination of up to 1,000 cells per fermenter at the end of the fermentation.

Additional data: the yeast growth rate is  $0.267 \text{ hr}^{-1}$  and the bacteria growth rate is 0.14 per hour.

**What is the minimal sterilization period required?**

*(10 min)*

### **Question 2**

A batch fermentation for growth of yeast cells is carried in a 1,500 liter fermenter. The medium contains molasses as the carbon source (69% solids while 58% of the solids is sugar), ammonia salts as the nitrogen source and Thiamine vitamin which is required in this fermentation. At the beginning of the fermentation the sugar concentration should be 40 gr/lit and the Thiamine concentration should be 200 ppm.

After medium preparation and before its inoculation the medium goes through a batch sterilization inside the fermenter. The sterilization is performed in a temperature of  $125^\circ\text{C}$  for 16 minutes (heating time of the medium to the sterilization temperature, as well as the cooling time can be neglected). The main source of contamination in the medium originates from the molasses, which contains spore concentration of  $10^7$  spores per gram molasses, and the sterilization process was planned so that the contamination probability will be only 1 to every 500 fermentations.

Additional data: the yeast growth yield is 0.674 gram yeast to gram molasses sugar, doubling time of yeast in this medium is 4455 seconds, the initial concentration of the yeast inoculated in the medium is 100 ppm, decomposition constant of Thiamine in the sterilization temperature is  $4.12 \text{ hr}^{-1}$ , the yeast start logarithmic growth immediately after inoculation.

**A. What is the death constant of the spores in the sterilization temperature?**

*(2.14 min<sup>-1</sup>)*

**B. Thiamine is provided to the medium as a concentrated stock solution with Thiamine concentration of 9%. What is the solution volume required to be added to the fermenter when preparing the medium?**

*(10 lit)*

**C. Assuming the molasses sugar are the limiting factor in the medium – how long will it takes for the fermentation to reach stationary phase?**

*(10 hr)*

The sterilization process is performed automatically, but due to a computer error, 6 hours after the fermentation begun, the fermenter sterilization process started again. Luckily, the operation engineer in the factory passed across the fermenter, noticed the problem and manually stopped the sterilization process after 5 minutes. At that stage, the medium was immediately cooled down to the fermentation temperature, and after a 1 hour lag stage, the yeast started again the logarithmic growth.

**D. Assuming the death coefficient of the yeast in the sterilization temperature is 0.35 per minute, what will be their concentration when they reach the stationary phase?**

*(17.62 grX/lit)*

**E. How long the fermentation will last from its beginning to the stationary phase?**

*(~13.4 hr)*

**F. What should be done to make sure that the yeast concentration at the end of this fermentation will be identical to their concentration at the end of a standard, normal fermentation? Detail all the assumptions and calculation required.**

### **Question 3**

A factory carries out batch fermentation of bacteria for the purpose of biomass production in a 500 liter fermenter. Each fermentation cycle takes 24 hr and includes: 6 hr of fermenter cleaning and filling for fermentation, 1 hr sterilization of the fermenter and its content, 1 hr of lag phase, 15 hr of logarithmic growth and 1 hr of fermenter discharge. The doubling time of the bacteria in this fermentation is 50.3 min and the growth yield on oxygen is  $1.2 \frac{\text{gr Bacteria}}{\text{gr Oxygen}}$ .

The air supplied to the system runs through continuous high temperature sterilizer ( $K_d=17.39 \text{ min}^{-1}$ ) on its way into the fermenter, while the staying time of the air in the continuous sterilizer is 88 sec. The initial concentration of contaminating cells in the air (before sterilization) is  $10^6$  cells per  $1 \text{ m}^3$  air, and the contamination risk level that the factory is willing to accept (after air sterilization) is 1 contamination per 1,000 fermentations.

During the fermentation, the oxygen percentage in the air which exits the fermenter is being continuously measured, and at the end of the fermentation the oxygen reading was 3.46%. It can be assumed that the air acts as ideal gas, in temperature of  $25^\circ\text{C}$  and pressure of 1 atmosphere.

- A. What is the bacteria concentration at the end of the fermentation?** (5gr/lit)
- B. If the cross section area (A) of the heat exchanger, where the air sterilization takes place, is  $73.35 \text{ cm}^2$ , what is the length of this heat exchanger (in meters)?** (25 m)